# Elliot Miller

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## Technical Skills

**Languages**: C++ (17-23), Rust, C, Assembly (x86-64/ARM), Dafny, Python, C#
**Libraries/Frameworks**: SDL, Tokio, AWS CDK/SDK, Unity
**Developer Tools**:Git, AWS, AWS Lambda, Docker, macOS, Linux, NVim

## Experience

**Software Engineering Intern (Rust)** — May 2026 – Aug 2026
*Apple (Cloud Elastic Disk - Control Plane)* — *Seattle, WA*

- Worked on a greenfield service offering low-latency cloud block storage for Apple's internal compute and ML platforms
- Designed and implemented a feature which allows platforms to recover data deleted in error within a grace period.

**Software Engineering Intern (Rust)** — Jan 2026 – May 2026
*N1* — *New York, NY*

- Contributed to a performant perpetual futures execution platform handling more than $2B in YTD volume and more than 100k trades per day at a startup backed by Founder's Fund
- Designed and implemented a high-performance rate limiter from scratch, inspecting every trader action (orders, deposits, etc.) between the NIC and matching engine layer to prevent DoS attacks
- Reduced the CPU usage of our deposit/withdrawal database by 98% (4 constantly pinned cores → 1 mostly idle core) by indexing expensive queries and optimizing views over large tables
- Mitigated a live production incident halting all deposits and withdrawals, saving over $50,000

## PROJECTS

GameBoy Emulator (C++23)

- Wrote a LR35902 Sharp CPU simulator/machine code interpreter capable of running GameBoy ROMs, emulating hardware bugs involving finicky instruction timing and interrupt handling
- Simulated all GameBoy-specific hardware components/systems, including internal clock circuits, interrupts, and proprietary graphics chip

Kademlia Implementation (Rust)

- Wrote a decentralized P2P file-sharing system in Rust featuring a custom-built DHT based on the Kademlia protocol inspired by the design of BitTorrent.
- Guarded against Sybil and Eclipse attacks by implementing SKademlia security extensions including an upgrade to SHA-2 and POW requirements to join the network

Snake Compiler (Rust, x86)

- Wrote a x86-64-emitting, optimizing compiler for a Python/OCaml hybrid language called Snake
- Implemented Rust interop, register allocation, dead code elimination, and dynamic typing, using lattices to eliminate unnecessary type checks

Formally Verified Sharded KV store (Dafny)

- Wrote a formally verified, distributed KV store in Dafny, proving safety properties for a partition-tolerant and available system

## Education

**University of Michigan** — Ann Arbor, MI
*Bachelor of Computer Science* — *Expected May 2027*

- Relevant Coursework: Formal Verification of Systems Software, Distributed Systems, Computer Networks, Operating Systems, Compilers, Web Systems, Computer Organization, Data Structures and Algorithms, Computer Theory, Game Development, Programming Languages